

09.04 The Unit Circle

Common problems and solution methods

Rule guide: recognize the task first

Recognize	Apply these rules
Trigonometric Functions And Models	Identify the angle measure, quadrant, or graph feature before selecting a formula.
The expression contains the chapter's key notation or model.	Write the known quantities, choose the matching formula, calculate carefully, and check the result in context.

Example 1: Identify the model

Problem

$$\sin 30^\circ = \frac{1}{2}$$

How to solve it

Write the known quantities and restrictions.

$$\sin \theta = \frac{\text{opposite}}{\text{hypotenuse}}$$

Check the result against the original statement.

Example 2: Apply the key rule

Problem

$$135^\circ \cdot \frac{\pi}{180}$$

How to solve it

Write the known quantities and restrictions.

$$B = \frac{2\pi}{T}$$

Check the result against the original statement.

Example 3: Check the result

Problem

$$y = 3\sin\left(\frac{\pi x}{2}\right) + 4$$

How to solve it

Write the known quantities and restrictions.

$$y = A\sin(Bx) + D$$

Check the result against the original statement.